Master 2 – Artificial Intelligence

Raphael Leonardi

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Deep Learning

- Neural Networks (CNN, RNN)
- Natural Language Processing
- Computer Vision
- Reinforcement Learning

Classical Machine Learning

- Random Forest, AdaBoost
- Gradient Boosting (XGBoost)
- Support Vector Machines (SVM)

Mathematics

- Linear Algebra
- Stats & Probability
- Analysis
- Error-correcting codes

LANGUAGES

French
Native

English
C1

PROGRAMMING LANGUAGES

Python · C / C++ · Java / C# · OCaml · SQL

Profile

Currently pursuing a Master's in Artificial Intelligence at Paris-Saclay University, with a focus on machine learning research — from deep learning architectures to the mathematical foundations behind them. Looking to bring that background into applied research or engineering roles.

Education

Master Artificial Intelligence — September 2025 - Currently
Paris-Saclay University

L3 — Computer Science Magistere — September 2024 - May 2025
Paris-Saclay University

L1-L2 — Double Bachelor's in Mathematics and Computer Science — September 2022 - May 2024
Paris-Saclay University

Internships & Professional Experience

Master 1 Internship — May 2026 - July 2026

Enhancing Extreme Rainfall Nowcasting with High-Resolution Lightning Data and Neural Networks

Laboratoire Sciences Pour l'Environnement | CNRS – Università di Corsica

- **Data Engineering:** Scaled high-throughput preprocessing pipelines across 26+ GB of NetCDF/sensor data on a computing cluster using Python and Pandas.
- **Multimodal Deep Learning:** Developed CNN-LSTM architectures with Multi-Head Attention to fuse Numerical Weather Predictions models, weather station feeds, and 3D lightning data.
- **Probabilistic Forecasting:** Implemented a mixed lognormal loss function to model full rainfall distributions and extreme precipitation events (1–6h horizons).
- **Evaluation & Fine-Tuning:** Assessed performance with Peirce Skill Score (PSS) and Negative Log-Likelihood, leveraging transfer learning and multi-seed validation.

Supervisor: Roberta Baggio

Magistere Internship — May 2025 - July 2025

Study of a post-quantum encryption protocol

INRIA Saclay — GRACE Team

- Discovery and understanding of error-correcting codes (Reed-Solomon, Reed-Muller)
- C implementation of the HQC post-quantum encryption algorithm
- Analysis of a scientific paper describing the implementation of an algorithm
- Exposure to presentations on advanced cryptography (elliptic curves, isogenies)

Supervisor: Christophe Levrat — christophe.levrat@math.cnrs.fr

Generative AI Expert in Freelance — January 2025 - Currently

Outlier

- Training of generative AI models
- Testing generative AIs and correcting contributions from other contributors